

Realistic Synthetic Data Generation for Defective Radiographic Testing Scans of Welds

Master's Thesis Topic Proposal — draft for discussion, August 2026

Bartu Can · Deeplify GmbH · bartu.can@deeplify.de · academic supervision sought at TUM

Motivation

Deeplify GmbH automates radiographic testing (RT) of welds: segmentation models localise six defect classes (porosity, inclusion, crack, undercut, lack of fusion/penetration) on digitised radiographs, trained on ~4,700 public-corpus films plus annotations from two industrial partners. Production experience exposes one dominant bottleneck: **annotated defective data is scarce, expensive, and extremely imbalanced**. The median defect covers 0.34% of a film's pixels; in a recent industrial annotation campaign only 23% of films contained any target defect. Per-class accuracy tracks per-class data volume (IoU ~0.73 for frequent classes vs. ~0.28 for rare ones), and domain gaps are severe — a model trained on the public corpus alone collapses on partner films (IoU 0.10 vs. 0.37 after adding partner data). A controllable source of realistic synthetic defective radiographs would attack all three problems at once.

Thesis Goal

Develop and evaluate generative methods that synthesise realistic defective RT scans — image plus pixel-accurate defect mask — such that training on real + synthetic data measurably improves downstream defect segmentation, especially for rare classes. The scientifically novel element is a **physics-informed formulation**: a radiograph follows Beer–Lambert attenuation ($I = I_0 \cdot \exp(-\int \mu \, dl)$), so in log-intensity space a real defect is an *additive, sign-constrained* perturbation determined by its thickness profile — not an arbitrary texture. Enforcing this as a constraint or loss during generation is, to our knowledge, unexplored for weld defects and is the core research gap.

State of the Art (to be surveyed in depth)

Physics-based simulators (aRTist [1], gVirtualXray) offer exact physics and free labels but suffer a sim-to-real appearance gap and need explicit defect geometry. **GAN-based defect synthesis** (Defect-GAN [2], few-shot DFMGAN [3]) is fast but unstable in exactly the low-data regime of rare defects. **Diffusion models** now dominate controllable synthesis; for industrial anomalies, AnomalyDiffusion [4] and RealNet [5] generate anomaly–mask pairs, and mask-conditioned *inpainting* [6, 7] allows injecting defects into abundant real defect-free films, inheriting authentic film appearance. **Cut-paste augmentation** [8] provides the cheap baseline every learned method must beat. None of these enforce radiographic physics — the proposed hybrid (physics-plausible attenuation perturbation + learned diffusion refinement [9, 10]) combines the strengths of the simulation and generative families.

Research Questions

RQ1 Which existing paradigm (simulation / GAN / diffusion inpainting / cut-paste) yields the most useful synthetic weld defects under one common evaluation?

RQ2 Can class- and mask-conditioned diffusion inpainting inject convincing defects into real defect-free radiographs? (~900 verified defect-free partner films are available as canvases; a trained weld-seam model, IoU 0.94, provides the placement prior.)

RQ3 Do physics-informed constraints — attenuation-consistency loss in log-intensity space, photon/grain noise realism, or generation in a thickness-perturbation space — improve realism and utility? Ablations isolate each term.

RQ4 Does it help downstream? Primary metric: per-class IoU and instance detection rate on a fixed, purely real validation set (real-only vs. real+synthetic at several synthetic fractions). Secondary: FID/KID and a blinded realism study with certified RT inspectors.

Setting and Work Plan

Deeplify provides the data (public corpus + growing partner annotations, 16-bit raws, verified defect-free films), a production training/evaluation stack with established baselines, GPU infrastructure, and access to certified inspectors — thesis results plug into an honest, fixed benchmark rather than a synthetic one. Preliminary 6-month plan:

M1–M2	Survey; cut-paste + GAN baselines; evaluation protocol; diffusion inpainting pipeline
M3–M4	Physics-informed formulation and ablations; scale generation; retrain and measure utility
M5–M6	Expert realism study; rare-class and domain-gap experiments; thesis writing

References (representative)

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- [10] G. E. Karniadakis et al.: Physics-Informed Machine Learning. Nature Reviews Physics, 2021.

Quantitative figures are Deeplify-internal measurements (August 2026); partners anonymised. Scope and methodology are open for discussion.